

# Hip — IA + IM

## Perinatal MSC therapy · intra-articular and intramuscular protocol

Hip pathology often involves both intra-articular structures (cartilage, labrum, synovium, capsule) and periarticular muscle (gluteal complex, hip flexors, adductors). This framework combines IA and IM delivery aimed at reducing inflammation and supporting cartilage and soft-tissue repair.

### INTRA-ARTICULAR

**40-100M**

viable MSCs in 3-5 mL carrier

### PER IM SITE

**10-20M**

across 2-3 sites, 1-2 mL aliquots

### TOTAL DOSE RANGE

**60-120M**

combined IA + IM, by patient size

### GUIDANCE

**US / fluoro**

strongly recommended

## Ideal candidates

- Adults 20-70 with MRI or radiographic evidence of hip pathology
- Mild-to-moderate hip OA (Kellgren-Lawrence II-III)
- Labral degeneration or tears not responsive to conservative care
- Chronic gluteal or adductor tendinopathy, muscle injury, post-surgical fibrosis
- Persistent pain or loss of function despite PT, NSAIDs, or prior biologics
- Motivated and able to comply with post-procedure rehabilitation

## Exclusions

- Advanced hip OA with bone-on-bone changes
- Active infection or systemic sepsis
- History of malignancy near the hip joint
- Severe uncontrolled autoimmune disease
- Pregnancy or breastfeeding
- Anticoagulation or coagulopathy unless corrected

## Pre-procedure preparation

- Informed consent explaining risks, benefits, and investigational status
- Discontinue NSAIDs  $\geq 5$  days pre-procedure
- Baseline pain score, range of motion, HOOS questionnaire
- Sterile environment; resuscitation equipment and monitoring available

**A · Intra-articular injection**

1. Patient supine; hip sterilized and draped.
2. Under fluoroscopy or ultrasound, identify anterior hip joint access.
3. Local anesthetic (1–2 mL lidocaine 1%) for skin and subcutaneous tissue.
4. Introduce spinal or long needle (22 G, 3.5–6 in by patient size) to the joint capsule.
5. Confirm intra-articular placement (saline aspiration or small contrast injection).
6. Inject 40–100M MSCs in 3–5 mL suspension slowly into the joint space.
7. Withdraw needle, apply sterile dressing.

**B · Intramuscular injections**

1. Lateral decubitus or prone depending on muscle group targeted.
2. Under ultrasound, identify target muscle (gluteus medius/minimus, piriformis, adductor longus).
3. Infiltrate local anesthetic at the skin entry point.
4. Advance 22–25 G spinal needle into the muscle belly.
5. Inject 10–20M MSCs per site in 1–2 mL aliquots.
6. Repeat at 2–3 sites depending on pathology.
7. Apply sterile dressing.

**Post-procedure protocol****IMMEDIATE****30–60 min observation**

Monitor for allergic response, vasovagal reaction, and injection-site complications.

**ACTIVITY****48 h rest · 1 week low impact**

No running or heavy lifting for one week. Resume physical therapy at 7–10 days: range of motion, then gradual strengthening.

**MEDICATION****No NSAIDs for 2 weeks**

NSAIDs may blunt MSC signaling. Acetaminophen, or a short opioid course, for pain if needed.

**Follow-up schedule****1 week**

Injection-site check, symptom monitoring

**1 month**

Pain and function assessment

**3 months**

Pain and function assessment, imaging if indicated

**6 months**

Durability review, repeat decision

**CLINICAL NOTE** Guidance modality, needle length, dose within range, and the number of IM sites remain physician decisions based on patient anatomy, imaging, and the pathology being addressed.

PATIENT NAME

DATE OF BIRTH

PATIENT ID / MRN

DATE OF VISIT

EVALUATING PHYSICIAN

INITIAL · FOLLOW-UP  
VISIT TYPE — CIRCLEREFERRING PROVIDER, IF  
ANY

TIME IN / OUT

**S SUBJECTIVE**

Hip / groin pain —

CHIEF COMPLAINT

ONSET / DURATION

SEVERITY — VAS 0-10

QUALITY / RADIATION

AGGRAVATING FACTORS

RELIEVING FACTORS

PT · NSAIDs · corticosteroid inj · activity modification · cane

PRIOR TREATMENT — CIRCLE ALL TRIED · ADD DATES

walking distance · stairs · shoes and socks · sitting tolerance · sl...  
FUNCTIONAL IMPACTfever · chills · weight loss · neuro changes — note positives  
PERTINENT ROSMEDICATIONS — INCL. ANTICOAGULANTS,  
IMMUNOSUPPRESSANTS

ALLERGIES · RELEVANT PMH — CANCER, INFECTION, PREGNANCY

**O OBJECTIVE**

BP

HR

TEMP

WEIGHT

GENERAL APPEARANCE

Gait — antalgia, Trendelenburg

Palpation — greater trochanter, groin

ROM — flexion / internal / external rotation, degrees

Provocative — FADIR, FABER

Neurovascular — distal pulses, sensation

IMAGING / STUDIES REVIEWED —  
MODALITY, DATE, FINDINGS

BASELINE — VAS /10

BASELINE — HOOS

BASELINE — Gait aid

**A ASSESSMENT**

Hip OA · degenerative hip pain · gluteal tendinopathy

WORKING DIAGNOSIS — CONFIRM OR AMEND

ADDITIONAL DIAGNOSES / CODES

☐ Appropriate candidate — indications met, contraindications  
absent☐ Alternatives discussed, including conservative care and doing  
nothing☐ Non-FDA-approved status explained; no outcome promised☐ Patient questions answered; understanding confirmed**P PLAN****Proposed treatment.** Image-guided intra-articular hip injection (perinatal MSC per plan, 4-6 cc total) with optional peritrochanteric or IM  
adjunct, 20-22 G, ultrasound or fluoroscopy. Observation 10-15 min.☐ Informed consent (CN-PC) reviewed and signed☐ Procedure performed today — see CN-OP-HIP · or scheduled (date  
right)☐ Fees, non-coverage by insurance, and refund policy reviewed☐ Follow-up: assessment at 4-6 wk; repeat consideration 6-12 mo

PHYSICIAN SIGNATURE

PRINTED NAME &amp; LICENSE NUMBER

PROCEDURE DATE, IF  
SCHEDULED

PATIENT NAME

DATE OF BIRTH

PATIENT ID / MRN

DATE OF PROCEDURE

PERFORMING PHYSICIAN

**Hip OA · degenerative hip pain · gluteal tendinopathy**

PRE-PROCEDURE DIAGNOSIS — CONFIRM OR AMEND

START TIME

ASSISTING

STAFF

**PRE-PROCEDURE CHECKLIST**

Check each box before the first draw

- |                                                                                                      |                                                                                                                             |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Informed consent (CN-PC) signed and on file; SOAP note (CN-SN-HIP) complete | <input type="checkbox"/> Screening reviewed — no contraindications present                                                  |
| <input type="checkbox"/> Time-out performed — correct patient, site, product, route                  | <input type="checkbox"/> Five checks read aloud with second staff: ID · product & lot · expiry · volume & route · consent   |
| <input type="checkbox"/> Baseline outcome measures recorded: VAS pain · HOOS · functional status     | <input type="checkbox"/> Site marked and prepped — chlorhexidine / povidone-iodine (circle); chain-of-custody log initiated |

BP · HR · TEMP

PRE-PROCEDURE VITALS

agent · concentration · volume — write NONE i...

LOCAL ANESTHETIC

BP · HR · TEMP

POST-OBSERVATION VITALS

**PRODUCTS ADMINISTERED**

Image guidance required for intra-articular hip placement

| PRODUCT & TARGET                                                           | PLANNED             | LOT | EXPIRY | GIVEN |
|----------------------------------------------------------------------------|---------------------|-----|--------|-------|
| <b>Perinatal MSC (per treatment plan)</b><br>Intra-articular, affected hip | <b>per plan</b>     |     |        |       |
| <b>Exosome adjunct (optional)</b><br>Peritrochanteric / IM, per plan       | <b>4-6 cc total</b> |     |        |       |

**TECHNIQUE**

Position: supine; anterior or anterolateral approach

- Confirm needle position under ultrasound or fluoroscopy before injecting.
- Aspirate before injection at every site; inject slowly.
- Optional peritrochanteric or IM component per treatment plan.

**LEFT · RIGHT · BILATERAL — circle**  
SITE / SIDE**ULTRASOUND · FLUOROSCOPY — required, circle**  
GUIDANCE**20-22 G, length per**  
**NEEDLE USED****target 10-15 min**  
**OBSERVATION — MINUTES**

NARRATIVE — FINDINGS, ASPIRATE, DEVIATIONS FROM PROTOCOL, PATIENT RESPONSE

**POST-PROCEDURE & DISPOSITION****Tolerated: WELL · WITH EVENTS — circle****Discharged: STABLE · AMBULATORY · WITH ESCORT — circle**

- ☐
- Patient note CN-PN-HIP reviewed and given at discharge

**COMPLICATIONS / ADVERSE EVENTS — WRITE "NONE" IF NONE**

Any adverse event also requires the adverse-event response steps in this book.

**Follow-up plan.** Clinical assessment at 4-6 weeks; repeat consideration at 6-12 months based on response.**FOLLOW-UP SCHEDULED FOR**

PHYSICIAN SIGNATURE

PRINTED NAME &amp; LICENSE NUMBER

DATE &amp; END TIME

Today you received an image-guided injection of cell products into your hip joint, with an optional injection near the outer hip. The products used are made from donated birth tissue and are not FDA-approved treatments. This page tells you what is normal in the days ahead, how to care for yourself, and when to call us.

DATE OF TREATMENT

SITE(S) TREATED

NEXT VISIT

**WHAT TO EXPECT**

Changes are gradual — do not judge the result in the first days

**FIRST 72 HOURS**

Deep soreness in the groin or outer hip is common and usually settles in two to three days. Walking may feel stiff at first.

**WEEKS 1-4**

Return to normal activity as comfort allows. Short, frequent walks are better than long outings early on.

**WEEKS 4-12**

Progress is assessed at your follow-up visit; improvement, if it comes, builds gradually over this window.

**DO**

- Rest the hip today; gentle walking around the house is fine
- Use cold packs over the injection site for comfort
- Take acetaminophen (Tylenol) for soreness if needed
- Resume physical therapy when advised

**AVOID**

- Anti-inflammatory pain relievers for 72 hours
- Alcohol for 72 hours
- Running, impact, and deep squats for 2-3 days
- Long car rides or prolonged sitting today if you can

**CALL US RIGHT AWAY IF**

For an emergency, call 911 first

- Increasing groin or hip pain with fever or chills
- Calf pain or swelling, or shortness of breath
- You cannot put weight on the leg
- Numbness or weakness in the leg that is new

CLINIC PHONE

AFTER-HOURS PHONE

**NOTES FROM YOUR VISIT**